

Research Engineer – Machine Learning & Robotics

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MSc in Robotics and Computer Engineering, working on diffusion-based video world models for robotics. Interested in scalable generative models, representation learning, and planning, with hands-on work in multi-GPU training and real-world robotic systems.

Master Thesis

Diffusion Video World Model for Robotic Planning

Oct 2025 - May 2026, Tartu, Estonia

- Developing a Video World Model on DINOv2 latent space, trained by Diffusion Forcing
- Built a unified data pipeline by converting large-scale ego-centric datasets (EPIC-KITCHENS, BridgeV2, SOAR, DROID) into LeRobotDataset format (Hugging Face)
- Distributed training (DDP) on the university HPC cluster using 8× NVIDIA H200 GPUs
- Implemented Transformer optimizations for stability and scale: RoPE, SwiGLU, RMSNorm and KV cache

Professional Experience

STMicroelectronics – Global leader in semiconductors

Grenoble, France

R&D Deep Learning Intern

4 months - 2025

- Developed a fully differentiable hybrid lithography simulator in PyTorch, combining a physics-based optical model (Hopkins) with a CNN for resist modeling
- Implemented an inverse-lithography (ILT) mask optimization pipeline using the hybrid simulator and gradient-based optimizers, for nanometer-scale hotspot reduction

Virtual Incision – Surgical robotics startup

Lincoln, USA

R&D Deep Learning Intern

4 months - 2024

- Developed Action-Chunking Transformer (ACT) in PyTorch, for surgical needle-picking and handoff via imitation learning on a dual-arm surgery robot
- Designed a real-time video upscaling GAN, implemented EDSR and ESRGAN papers (2K->4K)
- Implemented real-time inference deployment on NVIDIA Holoscan, reaching ~50ms latency
- Deployed a self-hosted data-labeling system (Label Studio + SAM) for surgical vision datasets

Michael Management – Freelance – Production of SAP training materials

Lyon, France - 2023

Restaurant La Cabane – Server

Chamonix Mont-Blanc, France - 2023

La Poste – Operations Intern

Lyon, France - 2022

Formations

Tartu University, Dual Degree – MSc in Robotics and Computer Engineering

Tartu, Estonia

Student Researcher at IMS Lab (Intelligent Materials & Systems)

Sept 2025 - May 2026

ECAM La Salle – Engineering degree

Lyon, France

MSc in Mechanical and Electrical Engineering (English program)

Sept 2021 - June 2026

Specialization in Robotics & Automation

Skills & Interests

- **ML** : Diffusion, Transformers, Computer Vision, PyTorch, NVIDIA Holoscan, multi-GPU training (DDP)
- **Programming** : Python, C/C++, Git, ROS, Arduino, Docker
- **Languages** : French (Native), English (Fluent - IELTS 7.5), Spanish (B1)
- **Interests** : Ski, Climbing, Swimming, Volley-ball, Keeping up with AI literature